EpiLink compatibility workflow

Observed testing dates and genetic distance are scored against latent transmission scenarios via Monte Carlo timing and genetic draws.

STEP 1

Observed pairwise data

Testing dates: $t_{test,i}$, $t_{test,j}$

Testing-time gap: t_{ij}

Consensus genetic distance: g_{ij}

STEP 2

Temporal model

Stages E → P → I with durations y_E, y_P, y_I

Testing delay: x_{test}

Generation time: τ_{gen}

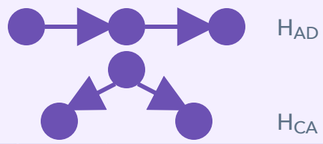
STEP 3

Candidate latent transmission scenarios

Ancestor-descendant: $H_{AD}(m)$

Common ancestor: $H_{CA}(m_i, m_j)$

m, m_i, and m_j denote hidden intermediates.



STEP 4

Monte Carlo temporal draws

For each scenario s, sample latent timings.

$$T_s = A_j(s) - A_i(s) + \tau_{inc,j} + x_{test,j} - \tau_{inc,i} - x_{test,i}$$

Store draws $T_s^{(1)}, \ldots, T_s^{(N)}$.

STEP 5

Monte Carlo branch-length draws

For each scenario s, compute branch-length draws.

$$AD: B_s = \Sigma\tau_{gen} + \tau_{inc,j} + x_{test,j} - (\tau_{inc,i} + x_{test,i})$$

$$CA: B_s = \Sigma\tau_{gen}^{(i)} + \Sigma\tau_{gen}^{(j)} + \tau_{inc,i} + x_{test,i} + \tau_{inc,j} + x_{test,j}$$

Store draws $B_s^{(1)}, \ldots, B_s^{(N)}$.

STEP 6

Mutation model

Strict clock: $\lambda = rL / 365$

Relaxed clock: $\lambda^{(n)}$ from UCLN draws

$$G_s = \lambda B_s$$

or Poisson(λB_s) for stochastic mutation counts.

STEP 7

Temporal compatibility

Compare observed t_{ij} with simulated T_s.

$$p_{T,s} = Pr(T_s \leq t_{ij})$$

$$C_{T,s} = 1 - 2|p_{T,s} - 0.5|$$

Highest when the observed value lies near the centre of the scenario-specific temporal distribution.

STEP 8

Genetic compatibility

Compare observed g_{ij} with simulated G_s.

$$p_{G,s} = Pr(G_s \leq g_{ij})$$

$$C_{G,s} = 1 - 2|p_{G,s} - 0.5|$$

Highest when the observed value lies near the centre of the scenario-specific genetic distribution.

STEP 9

Scenario and target scores

$$C_s(i,j) = C_{T,s}(i,j) \times C_{G,s}(i,j)$$

Compute this for every scenario $s \in S_M$.

$$score_{S^*}(i,j) = \Sigma C_s(i,j), \quad s \in S^*$$

A user-selected subset S* can represent one focal scenario or a broader family of scenarios.

Observed pair → profile and latent scenarios → Monte Carlo timing and genetics → scenario compatibilities → target score